

Statistical Models in Computational Biology

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An important driver of cancer is the mutation of tumor suppressor genes (TSG), among which we find genes coding for DNA repair proteins. The mutation and inactivation of such genes has the effect of increasing the frequency of mutations in a cell and its progeny. Suppose that we are investigating the inactivation of one specific TSG, and hypothesize that cells with this inactivation have a different probability of mutations than the ones without. The data we observe are sequencing results for certain positions of the genome, from a cell population at single cell resolution. We wish to find out from the data the effect of this TSG inactivation – how different are the probabilities of mutations with or without this inactivation? – as well as the proportion of cells with this inactivation in the population.

Problem 4: TSG inactivation problem formulation (5 points)

We model this problem with a two-component mixture model, corresponding to the cell groups with and without the TSG inactivation. Each component is parametrized by a mutation probability. Within a cell group, the probability of mutations are assumed to be the same at each position and the mutations are independent between positions.

More precisely, Let $\mathcal{D} = \{X^{(1)}, \dots, X^{(N)}\}$ be a set of sequences. Each $X^{(i)}$ has length L (we write $X^{(i)} = (X_1^{(i)}, \dots, X_L^{(i)})$), and $X_\ell^{(i)} \in \{0, 1\}$. In other words, we sequence N cells at L genomic positions, and represent the mutation status at each position by a binary variable, with 1 meaning mutated. Each sequence $X^{(i)}$ is generated either

- by the mutated model with parameter $\theta_1 \in (0, 1)$, where $P(X_\ell^{(i)} = 1) = \theta_1$, or
- by the wild-type model with parameter $\theta_2 \in (0, 1)$, where $P(X_\ell^{(i)} = 1) = \theta_2$.

The mutated model is used with probability λ_1 , and the wild-type model is used with probability $\lambda_2 = 1 - \lambda_1$.

We will learn maximum likelihood parameters $\lambda = (\lambda_1, \lambda_2)$ and $\theta = (\theta_1, \theta_2)$ using the EM algorithm.

- Let us define binary hidden variables $Z^{(ik)}$, where $Z^{(ik)} = 1$ if and only if $X^{(i)}$ is generated from component k . Write down the complete-data log-likelihood $\log P(X, Z \mid \lambda, \theta)$ in terms of $X^{(i)}$, $Z^{(i)}$, λ , and θ (i.e., do not use $P(\cdot)$ in the final answer). (2 points)
- Let us define the responsibility of component k for observation i as the posterior probability that $Z^{(ik)}$ is 1.

$$\gamma_{ik} := P(Z^{(ik)} = 1 \mid X, \lambda, \theta).$$

Express γ_{ik} in terms of $X^{(i)}$, λ , and θ . (1 point)

Hint: use Bayes' theorem.

- Derive the M-step update formulae for parameters λ and θ . (2 points)

Problem 5: TSG inactivation problem implementation**(5 points)**

Suppose that we sequenced 200 cells from a tumor biopsy sample. The file 'mutations.csv' contains mutation calls (denoted 1 if the position is mutated, and 0 otherwise) for 100 positions of interest in the genome, for the 200 cells in the sample. We will implement the EM algorithm formulated above, to learn which cells belong to which group and the mutation probabilities of each group.

Please use the R code skeleton `hw2_EM_skeleton.qmd` on Moodle for this question, and make sure to include the filled-out code sections in your submission.

- (a) Complete the implementation of the EM algorithm in the code skeleton. (2 points)
Note: Beware of numerical instability when multiplying large amount of numbers together – do additions of logarithms whenever possible.
- (b) Use the given code to read in the data stored in the file 'mutations.csv', and to initialize the mixture weights λ and the mutation probabilities θ . We assume a priori that the mixture weights are uniform, and $\theta_1 > \theta_2$. Run the completed EM algorithm on this data.
- (c) Plot the following quantities (y-axis) as a function of the number of iterations (x-axis) in three separate plots: (1 point)
- the observed log-likelihood (recall that this is computed by marginalizing out the hidden variables)
 - the mutation probabilities θ_1 and θ_2 , and
 - the mixture weights λ_1 and λ_2 .
- (d) Plot a heatmap of the responsibilities γ at the final step, where columns are sequences, rows are components, and values indicate the responsibility of component k for $X^{(i)}$. For each cell, we assign it to the component with higher probability. How many cells belong to the mutated group, and how many cells belong to the wild-type group? (2 points)